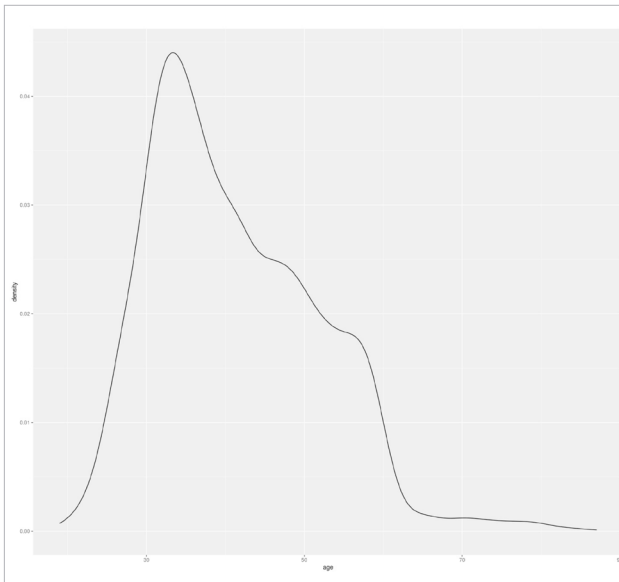


Figure 9: *ggplot* density graph

## ggplot

The *ggplot* function can be used to create a new *ggplot* object for input data, and also specify aesthetic mappings for the same.

For the *bank.csv* data, we can tabulate the job and marital status together using the *with* function as follows:

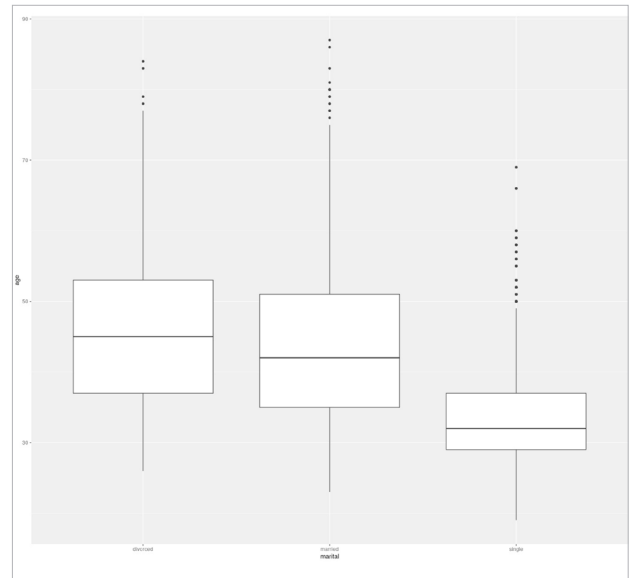
```
> with(bank, table(job, marital))
      marital
job      divorced married single
admin.         69      266    143
blue-collar    79     693    174
entrepreneur   16     132     20
housemaid      13      84     15
management    119     557    293
retired        43     176     11
self-employed  15     127     41
services       62     236    119
student        0       10     74
technician     89     411    268
unemployed     22      75     31
unknown        1       30      7
```

You can now plot the above categorical data using *ggplot*, as follows:

```
> ggplot(bank, aes(x = job, fill = marital)) + geom_bar()
```

The resultant graph is shown in Figure 8.

The age distribution can be plotted as a density using the *geom\_density* function as follows:

Figure 10: *ggplot* boxplot graph

```
> ggplot(bank, aes(x = age)) + geom_density()
```

The corresponding graph is shown in Figure 9.

A box plot for the age and marital status can be visualised using the following arguments to *ggplot*:

```
> ggplot(bank, aes(x = age, y = marital)) + geom_boxplot() +
coord_flip()
```

The output graph is as shown in Figure 10.

The *ggplot* function accepts the following arguments:

| Argument           | Description                                           |
|--------------------|-------------------------------------------------------|
| <i>data</i>        | The data frame for the plot                           |
| <i>mapping</i>     | The aesthetic mappings to be used in the plot         |
| <i>environment</i> | The <i>globalenv()</i> environment for the aesthetics |

Do try and explore more functions and charts in the graphics packages available in R. **END** 🐧

## References

- [1] India Inflation, consumer prices (annual %), <https://data.worldbank.org/indicator/FP.CPI.TOTL.ZG?locations=IN>
- [2] [Moro et al., 2014] S. Moro, P. Cortez and P. Rita. A Data-Driven Approach to Predict the Success of Bank Telemarketing. Decision Support Systems, Elsevier, 62:22-31, June 2014

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